

Finance.accdb — Phase 1 Discovery Report

Scope: database discovery only, per spec. No API design, schemas, or migration plans in this document.

Extraction method: direct page-level parse of the ACE database (access_parser, patched for ACE 2010 header), MSysObjects/MSysQueries/MSysRelationships system catalogs, and full VBA project extraction from MSysAccessStorage with MS-OVBA decompression. Raw evidence is in **extracted/** (all 110 VBA modules as .bas, all 313 reconstructed query SQL, full object inventory).

1. Database Inventory

Property	Value
File	Finance.accdb (36.7 MB)
Format	ACE 2010 (Access 2007–2016 era), page size 4096
Business	TIRUMALA FINANCE, Gajwel, Dist. Siddipet, Telangana (hardcoded in module T)
Vendor	"VNC Technologies" (hardcoded; includes software expiry lock in module SR)
Domain	Private lending: daily-interest loans (CD/LCD), monthly installment loans (STBD/ST/HP), daily installment loans (TBD), small savings (SS), partner capital accounts

Object type	Count
Tables (user)	20 (+1 attachment side-table, +25 system)
Queries	313 total: 130 named + 183 hidden ~sq_* (form/report/combo record sources)
Forms	107

Object type	Count
Reports	55
Standard VBA modules	13
Form/report code-behind modules	97 (110 VBA modules total, ~11,100 lines)
Macros	1 (~TMPCLPMacro, throwaway)
Relationships	0 user-defined

2. Tables

Core tables

Members — 528 rows, PK **Number** (text, e.g. "CD-17"). Despite the name, this is the **loan master**: one row per loan account. 58 columns. Key columns: **Date** (loan date), **Amount** (principal), **Period**, **Rate**, **Loantype**, **DueDate**, **Renewal** (bool), **RenewalDays**, **TotalLoan**, **ActualLoan**, **PaidAmount**, **OnlypremiumPaid** (principal-only paid), **InstallmentAmount**, **premiumamount**, **Idue/Ipaid** (installments due/paid), **DueAmount**, **duedays**, **Penalty**, **Balancewith/BalanceWithout** (balance with/without interest), **InterestAmount**, **LastDate**, **NPA**, **NPADATE**, **NPAAMOUNT**, guarantor fields ×2, **Partner**, **UserName**, **EntryTime**.

Day_book — 7,113 rows, PK **ID**. The **transaction ledger** (single-entry style journal). Key columns: **D_Date**, **Head_of_a/c** (account head), **Account_Number** (joins to Members.Number), **Particulars**, **Credit**, **Debit**, **DUEDATE** (carries the loan's next due date on payment rows), **RNO** (receipt no.), **LID** (loan id), **HC/PC/Commission/Penalty/FutureInterest** (charge components), **LP** (memo, penalty detail), **ENT**, **Direct**, **Closed**, **SS_ACNO**, **USerName**, **EntryTime**.

Customers — 353 rows, PK **ID**. Person registry (name, father, village/mandal/district, phones, Aadhaar).

Accounts — 59 rows, PK **Account_Name**. Chart of accounts; **Account_Type** from Account_types.

Lookup tables

Table	Rows	Content
Loan Types	4	STBD, CD, TBD, HP (VBA also branches on LCD and ST)
Account_types	7	Assets & Properties, Bank, Capital, In, LIABILITIES, Loans, Out
Usertypes	2	Administrator, User
Dates / Months / Years	31 / 12 / 91	Calendar pick lists
Company Name	1	Company display name/address

Audit / system-support tables

Table	Rows	Purpose
Deleted Daybook	819	Audit copy of deleted Day_book rows (+ user, delete time)
Edited Members	433	Before/after audit of Members edits (O*/N* column pairs)
Deleted Members	6	Audit of deleted loans
Backup	400	Backup file name log
Users	2	Login accounts — plaintext passwords
User2	1	Single shared/admin password
Partners_info	3	Partners (capital contributors), share details

Table	Rows	Purpose
Ledger	5	Scratch/temp installment schedule
ALL	0	Empty clone of Accounts (temp)

Classification: master = Members, Customers, Accounts, Partners_info; transaction = Day_book; lookup = Loan Types, Account_types, Usertypes, Dates, Months, Years, Company Name; audit = Deleted Daybook, Edited Members, Deleted Members, Backup; temp/scratch = Ledger, ALL; security = Users, User2.

3. Relationships

MSysRelationships contains zero user-defined relationships (only Access's internal nav-pane one). No referential integrity is enforced anywhere. All joins are by convention, recovered from the 313 queries:

Join (implicit FK)	Cardinality	Used for
Members.Number ↔ Day_book.Account_Number	1:N	Payment history, balance & due recomputation
Members.Number ↔ Day_book.SS_ACNO	1:N	Same link on charge rows
Members.Partner ↔ Partners_info.PartnerName	N:1	Partner performance/commission
Accounts.Account_Name ↔ Day_book.Head_of_a/c	N:1	GL grouping, P&L, FS
Accounts.Account_Type ↔ Account_types.Account_Type	N:1	FS/GL rollups
Members.Adhaar ↔ Customers.CAdhaar (and guarantor Aadhaar fields)	N:1	Customer 360 reports (loans, as guarantor 1/2)

Migration risk: orphan checks exist as queries ("Day_book Without Matching Members/Accounts") — the data is known to contain mismatches.

4. Queries

313 total. 130 named (full reconstructed SQL in [extracted/queries_named.txt](#)), 183 hidden `~sq_*` record sources ([extracted/queries_hidden.txt](#)). Functional groups:

Group	Representative queries
Payment aggregation	Loan Payments (sums Credit, takes <code>Max(DUEDATE)</code> per loan — feeds the recompute engine), Loan Payments L+S, Payments, daybookinterest (all "CD Commission A/c" rows)
Due/overdue lists	NPA QUERY, NPA LIST, Outstanding, Outstanding List, DUES DETAILS, DUES TOTAL, Partners Due List
Daily cash	DayBook, Daily Balances, Daily Closing Balance, Daily Report Query, Day Book Totals
P&L / financial statements	Profit Loss (1–3) series, Profit Loss BS series, FS*, GL* series, CAPITAL*
Partner commission	PARTNER COMMISSION (TOTAL), PARTNER PENALTIES, PARTNER DOCUMENT CHARGES, P Business*, PARTNER FINAL/GRAND TOTAL
Business volume	Business*, All Loans*, All Collection*, MDS BUSINESS*, This Month Opened/Closed Accounts
Data hygiene	Day_book Without Matching Accounts/Members, Members Without Matching PARTNERS_info, M for Edit, DayBook for Updating

Many queries take parameters from open forms (e.g. `Forms![PL Form]!From_Date`) — these dependencies must be cataloged in Phase 3.

5. Forms

107 forms (full list in `extracted/inventory.txt`). Functional map:

Cluster	Forms	Role
Login/shell	Login Form, Login Form Admin/User, Main Form for admin/User, RunningUser	Auth + navigation; RunningUser holds <code>CurrentDate</code>, the global business date used by all due/interest math
Loan origination	Loans Entry Form, MEMBER LOANS, Day Book Loans	Creates Members row + disbursement and upfront-charge Day_book postings
Loan servicing (per type)	CD LEDGER , CD LEDGER1, stbd ledger, STBD Daybook, TBD Ledger, SD Ledger, HP ledger, HP ledger1, Dues Ledger, Over Due List, NPA List	Payment receipt, renewal, closure — the business-logic heart
Cash book	Cash Book Entry Form (+ for Admin, Subform2), Daily Report	Non-loan receipts/payments
Parties	Customers, Partners, Partner * (7 forms), Phone/Adhaar forms	Registry maintenance
Savings/capital	SS ENTRY, SS CREATION, SS LEDGER, Capital Entry Form, Capital and SS Entry Form, CAPITAL DETAILS/TOTALS	Small savings and partner capital

Cluster	Forms	Role
Reconciliation/admin	Differences, Dif 1–6, NEW DIF, ED FORM, Edited Members, Deleted Members, FS, GL, PL Form, R Book, Find, WEBCAM, User*	Tools and reporting front-ends

6. Reports

55 reports (full list in [extracted/inventory.txt](#)):

Cluster	Reports
Collections/dues	Due List, CD DUE LIST, CD DUE LIST AB, Partners Due List, NPA LIST, Outstanding, new Outstanding, partners total outstanding
Daily operations	Daily Report for All Days / for Particular Day
Financial statements	FS, General Ledger, GL Detailed, GL GRAND TOTAL, GI Total, PL, PL Incomes/Expenses/New Loans/Closed Loans/All Loans/Present, Expenses
Partner	Partner Performance (+ Pwise, Receipts, DETAILS), partner performance DOCUMENT CHARGES / PENALTIES, Partner Names Id/Md/Name wise, MD Business Total
Customer 360	Customer Detailed/Total Report, Customer as G1/G2/Loans (+ totals), LoanPerson (+ G1Loans/G2Loans/Loans)
Business volume	Business, Business Detailed/Total/(Total), Business and Commission, PI All Collections
Documents	CD Ledger Sub Report (passbook/ledger print)

7. VBA Modules

13 standard modules + 97 form/report code-behind modules, all extracted to [extracted/vba_src/](#).

Standard module	Functions	Role
Updating Due Dates	UpdatingDueDate()	Recomputes Members.DueDate (see §8.1)
Updating Due List New	UpdatingDuelist() = FU1+FU2+FU3	Full loan-state recompute engine: FU1 re-derives loan constants by type; FU2 folds in payment aggregates; FU3 recomputes dues, balances, penalties (see §8)
Capital Updating	CapitalUpdating()	Rebuilds partner capital balances
Backup	Backuppingle, BackupFile, DeleteTheOldBackupFile	DB file backup rotation
Running Users and Current Date	CurrentDate(), PresentUserName()	Global business date + user (reads RunningUser form)
Amount Conversion in Words	Inwords()	Amount-to-words for receipts
SR	MYEXPDATE, WarningMessage, SoftwareRenewal	Vendor license expiry lock
T	CompanyName, CompanyAdd, OurTitle, Di	Hardcoded company strings
Updating Partner Ids in Members Table	UpdatingPartnerIds()	Data maintenance
Bypass	SetProperties(...)	Toggles Access bypass/admin properties
DDLK	DaLoc()	Dead code (commented-out date lock)

Standard module	Functions	Role
LP	4 functions, 3 empty	Mostly dead code

Key code-behind modules (heaviest logic): **Form_CD LEDGER** (renewal/closure engine), **Form_Loans Entry Form** (origination + upfront charges), **Form_stbd ledger**, **Form_TBD Ledger**, **Form_SD Ledger**, **Form_HP ledger**, **Form_Capital Entry Form**, **Form_Cash Book Entry Form**.

8. Where the four logic areas live

8.1 Due date calculation — found, HIGH confidence

Location	Logic
Updating Due Dates.bas → UpdatingDueDate()	Default: DueDate = Date + Period - 1 . CD loans: if Renewal=True → DueDate = Date + RenewalDays ; if Renewal=False → DueDate = Date + Period - 1
Form_Loans Entry Form (origination, lines 92–96, 672–678)	CD/LCD: DueDate = Date + Period (one path uses + Period - 1 — inconsistency to verify). HP/STBD/ST: DueDate = DateAdd("M", 1, Date) (first installment next month). TBD: DueDate = Date + 1
Updating Due List New.bas → FU3	Installment loans: DueDate = DateAdd("m", Round(Ipaid)+1, Date) ; TBD: DueDate = Date + Ipaid (days)
Form_CD LEDGER → RDAYS_AfterUpdate / RenBtn_GotFocus	On renewal: NextDueDate = DueDate + RDAYS — extends from the previous due date, not the payment date. Stored on the Day_book payment row (!DueDate = NextDueDate), then FU2 copies Max(DUEDATE) from payments back into Members.DueDate

Spec's example (loan 05-Feb-2022 + renewals 30+30+30+25+5 = 120 days → due date = 05-Feb-2022 + 120) is consistent with the chained **DueDate + RDAYS** model; exact verification against historical rows belongs to Phase 6.

8.2 Interest calculation — found, HIGH confidence

Per loan type:

Type	Formula	Source
CD/LCD (daily-interest demand loan)	$\text{Interest} = \text{Balancewith} \times \text{Rate}/100 \div 30 \times \text{Days}$ (Rate is monthly %, divided by 30 for daily; Days = days since DueDate or RDAYS being purchased). Upfront at origination: $\text{Credit} = \text{Amount} \times \text{Rate}/100/30 \times \text{Period}$ posted to "CD Commission A/c", plus doc/processing charges ($\text{CDHC} = \text{Amount} \times 3/100/30 \times \text{Period} / 2$, CDPC)	Form_CD LEDGER FindingDetails (lines 60–73), RDAYS_AfterUpdate; Form_Loans Entry Form lines 126–146; FU3 $(\text{DailyInterest} = \text{Balancewithout} \times \text{Rate}/100/30)$
STBD/ST (flat monthly installment)	$\text{TotalLoan} = \text{Amount} + \text{Amount} \times \text{Rate}/100 \times \text{Period};$ $\text{InstallmentAmount} = \text{Round}(\text{TotalLoan}/\text{Period});$ collection commission 3% × installments to "STBD Commission A/c"	FU1; Form_Loans Entry Form line 95; Form_stbd ledger Inst_Click
HP	Same flat model with hardcoded 2%/month in the recompute engine $(\text{TotalLoan} = \text{Amount} + \text{Amount} \times 2 \times \text{Period}/100)$ but entered Rate at origination — discrepancy to verify	FU1 vs Form_Loans Entry Form line 94

Type	Formula	Source
TBD (daily installment)	$\text{TotalLoan} = \text{Amount}$ (zero stated interest; income via upfront TBD PC charge)	FU1; Form_Loans Entry Form line 191

Penalty (CD): grace 5 days. $\text{DueDays} > 5 \rightarrow \text{Penalty} = \text{Balancewith} \times 0.75/100 \div 30 \times \text{pDays}$ where $\text{pDays} = \min(\text{RDAYS}, \text{DueDays})$; i.e. effective 0.75%/month surcharge on overdue days. Source: FindingDetails lines 66–72, RDAYS_AfterUpdate, CloseBtn_GotFocus.

8.3 Renewal calculation — found, HIGH confidence

Source: Form_CD LEDGER $\rightarrow$ RDAYS_AfterUpdate, RenBtn_GotFocus/Click, PartRenewal_Click; replayed by FU2/FU3.

Mechanics: operator enters renewal days (RDAYS) or an amount (TotalAmountPaying); the form converts money $\rightarrow$ days: $\text{RDAYS} = (\text{TotalAmountPaying} - \text{Penalty}) / \text{DailyInterest}$. $\text{Interest} = \text{Balancewith} \times \text{Rate}/100 \times \text{RDAYS}/30$. $\text{NextDueDate} = \text{DueDate} + \text{RDAYS}$. Postings (Day_book credits, all stamped with DueDate = NextDueDate): "CD Commission A/c" (interest, marked ENT=True, HC=credit, penalty memo in LP), "PENALTY CD A/C" (if any), and in part-renewal also "CD A/c" (principal). Principal is NOT reduced by interest-only renewal; Renewal/RenevalDays flags on Members feed UpdatingDueDate(). Renewal precondition (interest must be paid) is enforced by the money $\rightarrow$ days conversion itself; FU2 then syncs Members from the latest payment row.

8.4 Payment allocation — found, HIGH confidence

Order confirmed in code: **Penalty** $\rightarrow$ **Interest** $\rightarrow$ **Principal**.

Evidence	Detail
Form_CD LEDGER.RenBtn_GotFocus	From TotalAmountPaying: penalty deducted first ($\text{RDAYS} = (\text{Paying} - \text{Penalty}) / \text{DailyInterest}$), remainder buys interest days; $\text{Penalty} = \text{Paying} - \text{Interest}$ recheck, floor 0. No principal component in plain renewal
Form_CD LEDGER.PartRenewal_Click	Three ordered postings: principal ("CD A/c"), interest ("CD Commission A/c"), penalty ("PENALTY CD A/C") — each a separate Day_book row keyed by RNO

Evidence	Detail
Form_CD LEDGER.CloseBtn_GotFocus / Command217 (NPA close)	Closure = Balancewith + Interest + Penalty; NPA close posts principal as "NPA CLOSE"
Form_stbd ledger.Inst_Click	Installment receipt splits into: principal ($\text{Amount}/\text{Period} \times n \rightarrow \text{"STBD A/c"}\text{"}$), commission ($\text{Amount} \times 3\% \times n \rightarrow \text{"STBD Commission A/c"}\text{"}$), penalty $\rightarrow \text{"STBD PENALTY A/c"}\text{"}$
FU2/FU3	Only "CD Commission A/c" rows advance DueDate without touching principal; other credits increase OnlypremiumPaid (principal repaid)

Notable observations for later phases (not analyzed further per scope)

Hardcoded constants: CD penalty 0.75%/month, 5-day grace, HP 2%/month in recompute engine, STBD commission 3%, CD doc charge $3\%/100/30 \times \text{Period}/2$, NPA window $\text{lastdate} = \text{Date} + 180$, photo paths `D:\Photos\`. Global business date comes from an open form (`Forms!RunningUser!CurrentDate`), not the system clock. Plaintext passwords in Users/User2. Zero referential integrity; known orphan rows. One inconsistency to resolve: $+\text{Period}$ vs $+\text{Period} - 1$ in CD due date between origination paths. Vendor license-lock module (SR) must not be migrated.

Stopped after discovery as instructed. Next steps when you're ready: Phase 2–9 deep dives (formula verification against historical Day_book rows, query-by-query documentation, workflow mapping).